

Digital Filter Design Using Genetic Algorithm

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Abstract: In this paper, the authors present an algorithm for designing 1-D FIR filters using Genetic Algorithms. In addition, the authors examine the usefulness of various error norms such as I_2 and I_∞ (minimax), and their impact on convergence rate and optimal result. The authors also present the application of various coding schemes applied to the filter coefficients and their effect in obtaining optimum filters. Examples of 1-D FIR filters are given using the above methodologies to illustrate the usefulness and utilities of the proposed techniques.

1. INTRODUCTION

The implementation of digital filters using infinite-precision coefficients introduces truncation and quantization errors. This reduces the filter performance due to the modifications of the coefficients. The utilization of discrete form with finite wordlength coefficients has the advantage of eliminating these problems.

Recently, Genetic Algorithm (GA) [5] has been emerged into optimum filter designs [1], [2]. The characteristics of multi-objective, coded variables, and natural selection make GA different from other optimization techniques [5]. Filters designed by GA have the potential of obtaining near global optimum solution. By utilizing the characteristic of GA encoding scheme, resultant filter coefficients are in discrete form without quantization process. Earlier works by D. Suckley [1], R. Cemes, D. Ait-Boudaoud [2], G. Wade, A. Roberts, G. Williams [3] propose 1-D FIR filter designs using power-of-two (POT) coefficients which result in multiplier-less digital filters.

As an extension of POT filter design, we propose a modified genetic algorithm (GA) structure to design digital FIR filters. The resulting filter coefficients are guaranteed to be in canonical signed-digits (CSD) format. By allowing more than one non-zero POT digits present in each of the coefficients, designed filters will have better performance, and the filter implementations are less complex since full size multipliers are not used in this structure.

In this paper, we emphasize on the design of FIR filters using genetic algorithm. The next section is a brief review of genetic algorithm. The remaining sections will explain

different encoding, and parent selection schemes in designing filters. Finally, the results of various design methodologies are compared.

Digital FIR Filter Design

A digital FIR filter is characterized by the following transfer function:

$$H(z) = \sum_{n=0}^N a_n z^{-n},$$

where N is the order of the filter, and a_n is defined as the filter coefficients. By determining the values of a_n , different types of filters can be implemented, such as lowpass, highpass, and bandpass filters.

2. INTRODUCTION OF GENETIC ALGORITHM

The genetic algorithm is an artificial genetic system based on the processes of natural selection, and natural genetic. Basic GA involves three operators: reproduction, crossover, and mutation. A simplified GA cycle is shown in Fig.1.

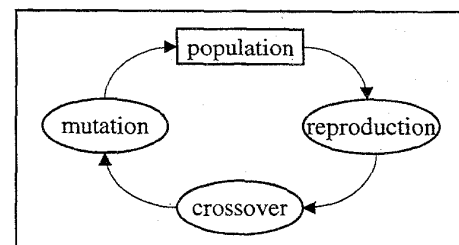
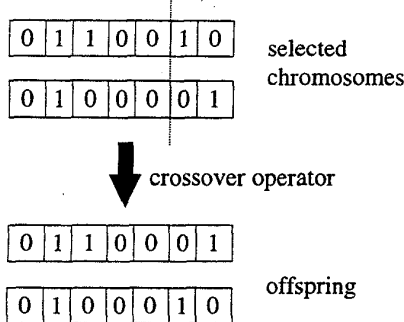


Fig.1 GA cycle

Population is a collection of chromosomes. Each of the chromosomes is a bit stream of cascaded variables (encoded filter coefficients). For a given *parent selection scheme* within reproduction operator, more suitable chromosomes are selected from the population for further genetic enhancements. A chromosome is *more suitable* if it results in lower cost-function error. Selected chromosomes are sent

to the next operator, crossover. In this stage, the GA algorithm randomly selects a common point in the chromosomes and exchanges their corresponding digits.



In the above example, two chromosomes are chosen from a population. The right side of the dotted line shows the binary digits which participate in this process. Two new chromosomes, called offspring, are created by exchanging their last two digits. Each of the offspring corresponds to a set of newly created variables. The last operator is mutation. Every bit contained in the offspring will be examined in this stage. Within a specified probability, certain digits will be altered from either '0' to '1' or '1' to '0' in binary encoding.

3. FILTER COEFFICIENT ENCODING

One of the characteristics of GA is the direct manipulation on coded variables which provides flexibility to solve different optimization problems. We consider two types of encoding schemes, ternary and mixed encoding, to represent filter coefficients.

3.1 Ternary Encoding

In this encoding scheme, a filter coefficient is represented by a string of ternary digits. Each of the ternary digits corresponds to a power-of-two (POT) number. Consider a POT number, $k \cdot 2^m$. If k is positive (+1), position $|m|$ of the string is 1. If k is negative (-1), position $|m|$ of the string is shown by $\bar{1}$. Otherwise, the digit is zero. As an example, consider 0.421875 in ternary encoding.

$$\text{coefficient } 0.421875 \Rightarrow 2^{-1} - 2^{-4} - 2^{-6} \\ \Rightarrow \begin{array}{|c|c|c|c|c|c|c|} \hline 0 & 1 & 0 & 0 & \bar{1} & 0 & \bar{1} & 0 \\ \hline \end{array} \quad \text{LSB}$$

The ternary string length is equivalent to the wordlength of filter coefficient. Cascading more than one encoded coefficient will form a row of ternary bit stream which is a chromosome in GA.

3.2 Mixed Encoding

In this encoding scheme, each filter coefficient is represented by the combination of ternary digits and binary strings. Consider power-of-two (POT) numbers, $k_1 \cdot 2^{m_1}$, $k_2 \cdot 2^{m_2}$, and $k_3 \cdot 2^{m_3}$. In mixed encoding scheme, ternary value k_i , which is 1, 0, or -1, is followed by a binary representation of m_i . In other words, the ternary digits which represent the sign, followed by the binary strings, which represent the exponent of the POT numbers form the code. Consider similar example, 0.421875.

$$\text{coefficient } 0.421875 \Rightarrow 2^{-1} - 2^{-4} - 2^{-6} \\ \Rightarrow \begin{array}{|c|c|c|c|c|c|c|c|} \hline 1 & 0 & 0 & 1 & \bar{1} & 1 & 0 & 0 & \bar{1} & 1 & 1 & 0 \\ \hline \end{array} \quad \begin{array}{l} \text{MSB} \\ \text{LSB} \end{array}$$

In the above example, a filter coefficient is written as the sum of power-of-two numbers, 2^{-1} , -2^{-4} , and -2^{-6} . For illustration purposes, the -2^{-6} corresponds to the last two blocks in which the sign and the exponent are represented by a $\bar{1}$ and a binary string of {1 1 0} respectively. A {0} ternary digit implies an invalid corresponding binary string. The wordlength of a filter coefficient is bounded by the maximum value of the exponent, $(m_i + 1)$. In this example, three bits binary strings can be used to encode the filter coefficient for maximum wordlength of 8 bits. A chromosome can be constructed by cascading these coded coefficients. Only non-zero POT digits are encoded for mixed encoding, whereas the whole filter coefficient including the zero digits is encoded for ternary encoding.

4. PARENT SELECTION

In selecting more suitable chromosomes from a population, we consider two types of parent selection schemes. The first one is the commonly used roulette wheel method [4], and the second one is modified version of the ranking parent selection [5].

4.1 Roulette Wheel Parent Selection

The basic roulette wheel parent selection is taken from a game of roulette wheel. It intends to assign a higher probability to those (chromosomes) which associate with a relatively high fitness value among the population. It involves the following four steps:

1. Add all the fitness values.
2. Randomly select a number between zero and the sum of fitness values.
3. Return the first chromosome that the sum of all its preceding fitness values are greater than or equal to the randomly selected number.
4. Go back to step1 until N_{chrom} number of chromosomes are selected, where N_{chrom} is the population size.

4.2 Modified Ranking Parent Selection

Under this selection scheme, similar fitness chromosomes are grouped in pairs. This may enhance the overall convergence rate at the beginning of the iterations. The convergence rate decreases as it approaches to an optimum solution. This scheme involves the following two steps.

1. Rank the chromosomes in descending order according to their fitness values, eg. [chrom1 chrom2 chrom3 chrom4 ... chromN-1 chromN].
2. Group the ranked chromosomes in pairs, and pass them over to crossover operation, eg. [(chrom1, chrom2), (chrom3, chrom4),, (chromN-1, chromN)].

Modified ranking parent selection aims only at the local optimum search rather than a global solution. Since GA is a stochastic search technique, more than one trial is necessary to obtain better solutions.

5. DESIGN FORMULATION

Minimax and least mean square (LMS) strategies are used to evaluate the fitness of chromosomes (filter coefficients). Digital FIR filter transfer function is defined as

$$H(z, \varphi) = \sum_{n=0}^N a_n z^{-n},$$

where φ is the vector of filter coefficients, $[a_0 \ a_1 \ a_2 \ \dots \ a_{N-1} \ a_N]$, and 'N' is the order of the filter. The corresponding frequency response is obtained by substituting $z = e^{j\omega T}$,

$$H(\omega, \varphi) = \sum_{n=0}^N a_n e^{-jn\omega T},$$

where T and ω are the sampling period and angular frequency respectively.

5.1 Minimax Error

Minimax objective function is defined as follows. It searches for the maximum peak error throughout a discrete frequency domain by subtracting the magnitude response of the ideal filter to designed filter,

$$Error = E_{\ell_{\infty}} = \max \left| |H_I(\omega)| - |H_D(\omega, \varphi)| \right|,$$

where $H_I(\omega)$ and $H_D(\omega, \varphi)$ are defined as the frequency responses of the ideal and designed filters.

5.2 Least Mean Square (LMS) Error

Least mean square error objective function is defined as the sum of the errors throughout a discrete frequency domain by

subtracting the magnitude response of the ideal filter to designed filter,

$$Error = E_{\ell_2} = \left\{ \sum_{\omega} \left[|H_I(\omega)| - |H_D(\omega, \varphi)| \right]^2 \right\}^{\frac{1}{2}}.$$

6. MODIFIED GA STRUCTURE

A modified GA structure is used to compensate the canonical signed-digit (CSD) constraints. These constraints are imposed by filter coefficients, and they are explained as follows (see [6] for more details of CSD):

1. The number of non-zero digits within a coefficient must be less than or equal to a prescribed number, N_{digits} .
2. The adjacent non-zero digit must be a '0'.

A modified GA structure is shown in Fig.2. The block diagram contains basic GA operators (reproduction, crossover, mutation), and three additional processes (CSD check, max. failure, reset coefficient) outlined by the dotted line. These additional processes, explained next, will guarantee filter coefficients to be of CSD format.

6.1 CSD check

Each newly created bit stream (offspring) is examined to see if the individual coefficients satisfy the above CSD constraints. Any violation will cause the bit stream (offspring) to be considered as 'failed'.

6.2 Max. failure

For a specified maximum number of failures from 'CSD check', N_{max} , individual coefficients can 'fail' no more than N_{max} times. If one of the individual coefficient has failed for N_{max} times, it is directed to 'Reset coefficient' module as in Fig.2. Otherwise, this pair of coefficients will go through crossover and mutation again.

6.3 Reset coefficient

The incoming coefficient will be disregarded, and replaced by a randomly generated non-zero power-of-two digit according to its encoding scheme.

7. SIMULATION RESULTS

Different combinations of encoding schemes, parent selections, and cost-functions are utilized to design FIR filters. In order to further enhance the GA performance, a multi-point crossover is used, such that more than one coefficient of selected chromosomes will experience crossover simultaneously.

The functionality of 'initialization' in Fig.2 is to create a population of size N_{chrom} by selecting the bits randomly. Each chromosome within a population is cascaded of encoded filter coefficients, such that each coefficient is initialized by a non-zero power-of-two digit.

'Get fitness' in Fig.2 evaluates the chromosomes (or offspring) by either the minimax or LMS strategy. The reciprocal of the evaluated error is stored along with its corresponding chromosome (or offspring) using

$$fitness = \frac{1}{Error}.$$

7.1 Discussion

Various orders of lowpass filters were designed, and their performance were compared on different encoding schemes, parent selection schemes, and the usage of cost-functions.

7.1.1 Cost-Functions Comparison

Fig.3 shows the error versus the number of coefficients using two different cost-functions. We have concluded that minimax strategy generates better results than LMS for all combinations on encoding schemes and parent selections.

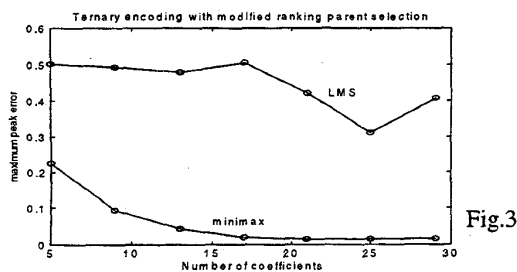


Fig.3

7.1.2 Parent Selection Schemes Comparison

Fig.4 and 5 show the performance of filters designed for different parent selection schemes in terms of errors and convergence rate. We have concluded that their performances are similar.

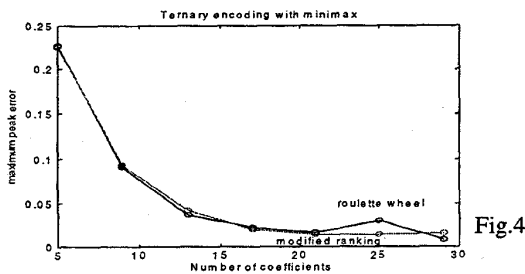


Fig.4

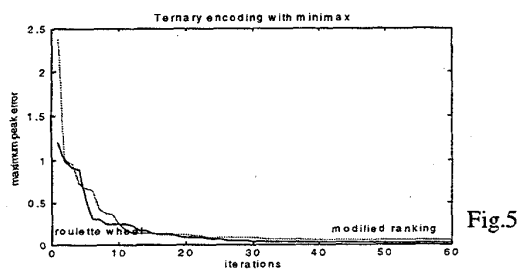


Fig.5

7.1.3 Filter coefficient Encoding Schemes Comparison

Fig.6 shows that the designed filters have similar performances regardless of which encoding scheme is used: ternary, or mixed encoding.

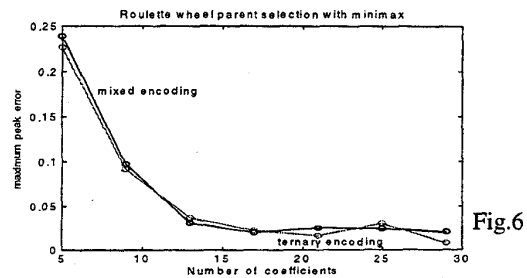


Fig.6

However, ternary encoding provides a faster convergence rate than mixed encoding as shown in Fig.7. Such indication is not conclusive when modified ranking parent selection is used.

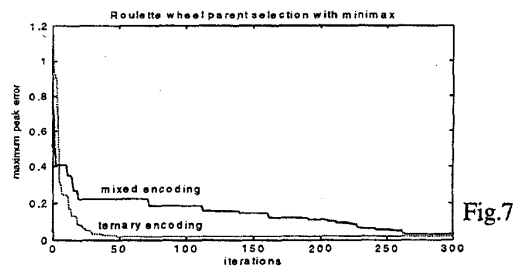


Fig.7

In practice, fewer iterations are required to obtain better results if we use ternary encoding. On the other hand, mixed encoding generally produces chromosomes with shorter length.

7.2 Example

As an example, a 28th order FIR filter is designed using ternary encoding with roulette wheel parent selection scheme. Its magnitude response and the corresponding filter coefficients are shown in Fig.8 and Table 1.

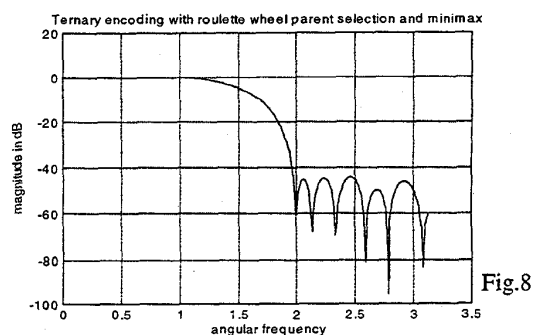


Fig.8

8. CONCLUSION

In this paper, we have presented an iterative method to design digital FIR filter using genetic algorithm. Additional processes are used to ensure that the filter coefficients are canonical signed-digits. We have shown that GA is robust in both encoding and parent selection schemes. The combination of encoding with either modified ranking or roulette wheel method produce similar results. Minimax strategy generally produces better filters than LMS. The modified GA structure presented in this paper can be extended to solve different kind of constraints problems.

References:

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coefficients	value
a(0)	-2^{-10}
a(1)	-2^{-8}
a(2)	$2^{-9} + 2^{-11}$
a(3)	$2^{-8} + 2^{-11}$
a(4)	$-2^{-9} - 2^{-11}$
a(5)	$-2^{-9} + 2^{-11}$
a(6)	$2^{-9} - 2^{-11}$
a(7)	$-2^{-7} + 2^{-10}$
a(8)	2^{-9}
a(9)	$2^{-5} - 2^{-9} + 2^{-11}$
a(10)	$-2^{-7} + 2^{-9} + 2^{-11}$
a(11)	$-2^{-4} - 2^{-6} - 2^{-8}$
a(12)	$2^{-7} + 2^{-10}$
a(13)	$2^{-2} + 2^{-4} - 2^{-9}$
a(14)	$2^{-1} - 2^{-7} - 2^{-9}$
a(15)	$2^{-2} + 2^{-4} - 2^{-9}$
a(16)	$2^{-7} + 2^{-10}$
a(17)	$-2^{-4} - 2^{-6} - 2^{-8}$
a(18)	$-2^{-7} + 2^{-9} + 2^{-11}$
a(19)	$2^{-5} - 2^{-9} + 2^{-11}$

a(20)	2^{-9}
a(21)	$-2^{-7} + 2^{-10}$
a(22)	$2^{-9} - 2^{-11}$
a(23)	$-2^{-9} + 2^{-11}$
a(24)	$-2^{-9} - 2^{-11}$
a(25)	$2^{-8} + 2^{-11}$
a(26)	$2^{-9} + 2^{-11}$
a(27)	-2^{-8}
a(28)	-2^{-10}

TABLE 1: 28th order FIR filter coefficients

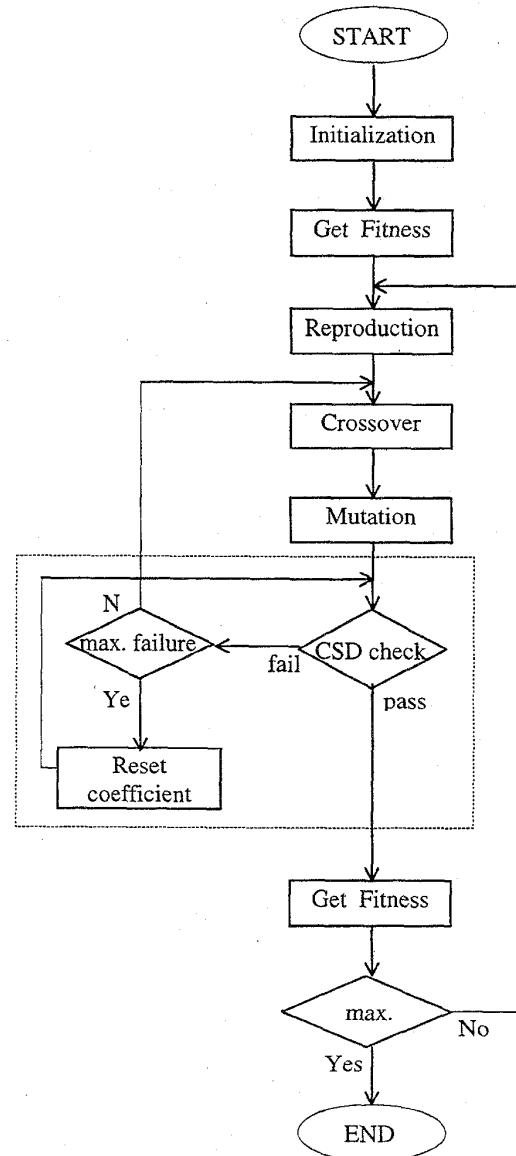


Fig.2 modified GA structure